Linear Equations and Determinants

Determinants and Cramer's Rule

Determinants

- If a matrix is square (that is, if it has the same number of rows as columns), then we can assign to it a number called its determinant.
  - *Determinants can be used to solve systems of linear equations--as we will see later in the section.*
  - *They are also useful in determining whether a matrix has an inverse.*

Determinant of 1 × 1 matrix

- We denote the determinant of a square matrix A by the symbol $\det(A)$ or $|A|$.

- We first define $\det(A)$ for the simplest cases.
  - If $A = [a]$ is a 1×1 matrix, then $\det(A) = a$.

Determinant of a 2 × 2 Matrix

The determinant of the 2 × 2 matrix is:

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

$$\det(A) = |A| = \begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$$

Determinant of an n × n Matrix

- To define the concept of determinant for an arbitrary $n \times n$ matrix, we need the following terminology.

Let A be an $n \times n$ matrix:

- The minor M_{ij} of the element a_{ij} is the determinant of the matrix obtained by deleting the i_{th} row and j_{th} column of A.

- The cofactor A_{ij} of the element a_{ij} is:

$$A_{ij} = (-1)^{i+j} M_{ij}$$

Determinant of an n × n Matrix

- For example, A is the matrix

$$\begin{bmatrix} 2 & 3 & -1 \\ 0 & 2 & 4 \\ -2 & 5 & 6 \end{bmatrix}$$

  - The minor M_{12} is the determinant of the matrix obtained by deleting the first row and second column from A.

$$M_{12} = \begin{vmatrix} 2 & 3 & 1 \\ 0 & 2 & 4 \\ -2 & 5 & 6 \end{vmatrix} = \begin{vmatrix} 0 & 4 \\ -2 & 6 \end{vmatrix} = 0(6) - 4(-2) = 8$$

  - So, the cofactor $A_{12} = (-1)^{1+2}M_{12} = -8$

Determinant of an n × n Matrix

- Similarly,

$$M_{33} = \begin{vmatrix} 2 & 3 & -1 \\ 0 & 2 & 4 \\ -2 & 5 & 6 \end{vmatrix} = \begin{vmatrix} 2 & 3 \\ 0 & 2 \end{vmatrix} = 2 \cdot 2 - 3 \cdot 0 = 4$$

- So, $A_{33} = (-1)^{3+3} M_{33} = 4$

Determinant of an n × n Matrix

- Note that the cofactor of a_{ij} is simply the minor of a_{ij} multiplied by either 1 or –1, depending on whether $i + j$ is even or odd.
  - Thus, in a 3×3 matrix, we obtain the cofactor of any element by prefixing its minor with the sign obtained from the following checkerboard pattern.

$$\begin{bmatrix} + & - & + \\ - & + & - \\ + & - & + \end{bmatrix}$$

Determinant of a Square Matrix

- If A is an $n \times n$ matrix, the determinant of A is obtained by multiplying each element of the first row by its cofactor, and then adding the results.

$$\det(A) = |A| = \begin{vmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{vmatrix}$$

$$= a_{11}A_{11} + a_{12}A_{12} + \ldots + a_{1n}A_{1n}$$

Determinant of a 3×3 Matrix

Example-1

Evaluate the determinant of the matrix.

$$A = \begin{bmatrix} 2 & 3 & -1 \\ 0 & 2 & 4 \\ -2 & 5 & 6 \end{bmatrix}$$

Determinant of a 3×3 Matrix

$$\det(A)$$

$$= \begin{vmatrix} 2 & 3 & -1 \\ 0 & 2 & 4 \\ -2 & 5 & 6 \end{vmatrix}$$

$$= 2\begin{vmatrix} 2 & 4 \\ 5 & 6 \end{vmatrix} - 3\begin{vmatrix} 0 & 4 \\ -2 & 6 \end{vmatrix} + (-1)\begin{vmatrix} 0 & 2 \\ -2 & 5 \end{vmatrix}$$

$$= 2(2 \cdot 6 - 4 \cdot 5) - 3\left[0 \cdot 6 - 4(-2)\right] - \left[0 \cdot 5 - 2(-2)\right]$$

$$= -16 - 24 - 4$$

$$= -44$$

Expanding the Determinant

- In our definition of the determinant, we used the cofactors of elements in the first row only.
  - This is called expanding the determinant by the first row.
  - In fact, we can expand the determinant by any row or column in the same way, and obtain the same result in each case.
  - We won’t prove this, though.

Expanding the Determinant

<u>Example 1. (a)</u>

- Expanding by the second row, we get:

$$\det(A) = \begin{vmatrix} 2 & 3 & -1 \\ 0 & 2 & -4 \\ -2 & 5 & 6 \end{vmatrix}$$

$$= -0\begin{vmatrix} 3 & -1 \\ 5 & 6 \end{vmatrix} + 2\begin{vmatrix} 2 & -1 \\ -2 & 6 \end{vmatrix} - 4\begin{vmatrix} 2 & 3 \\ -2 & 5 \end{vmatrix}$$

$$= 0 + 2\left[2 \cdot 6 - (-1)(-2)\right] - 4\left[2.5 - 3(-2)\right]$$

$$= 0 + 20 - 64 = -44$$

Expanding the Determinant

<u>Example 1.(b)</u>

- Expanding by the third column, we get:

$$\det(A) = \begin{vmatrix} 2 & 3 & -1 \\ 0 & 2 & 4 \\ -2 & 5 & 6 \end{vmatrix}$$

$$= -1\begin{vmatrix} 0 & 2 \\ -2 & 5 \end{vmatrix} - 4\begin{vmatrix} 2 & 3 \\ -2 & 5 \end{vmatrix} + 6\begin{vmatrix} 2 & 3 \\ 0 & 2 \end{vmatrix}$$

$$= -\left[0 \cdot 5 - 2(-2)\right] - 4\left[2 \cdot 5 - 3(-2)\right]$$

$$+6\left[2.2 - 3 \cdot 0\right]$$

$$= -4 - 64 + 24 = -44$$

Expanding Determinant about Row and Column

In both cases, we obtain the same value for the determinant as when we expanded by the first row in Example 1.

Inverse of Square Matrix

- The following criterion allows us to determine whether a square matrix is invertible or not without actually calculating the inverse.
  - This is one of the most important uses of the determinant in matrix algebra.
  - It is reason for the name determinant.

Invertibility Criterion

- If A is a square matrix, then A has an inverse if and only if $\det(A) \neq 0$.
  - We will not prove this fact.
  - However, from the formula for the inverse of a 2×2 matrix, you can see why it is true in the 2×2 case.

Determinant to Show Matrix Is Not Invertible

- Show that the matrix A is not invertible.
  - We begin by calculating the determinant of A.
  - Since all but one of the elements of the second row is zero, we expand the determinant by the second row.

$$A = \begin{bmatrix} 1 & 2 & 0 & 4 \\ 0 & 0 & 0 & 3 \\ 5 & 6 & 2 & 6 \\ 2 & 4 & 0 & 9 \end{bmatrix}$$

Determinant to Show Matrix Is Not Invertible

If we do so, we see from this equation that only the cofactor A_{24} needs to be calculated.

$$\det(A)$$

$$= \begin{vmatrix} 1 & 2 & 0 & 4 \\ 0 & 0 & 0 & 3 \\ 5 & 6 & 2 & 6 \\ 2 & 4 & 0 & 9 \end{vmatrix}$$

$$= -0 \cdot A_{21} + 0 \cdot A_{22} - 0 \cdot A_{23} + 3 \cdot A_{24}$$

$$= 3A_{24}$$

Determinant to Show Matrix Is Not Invertible

$$
\begin{aligned}
&= 3\begin{vmatrix} 1 & 2 & 0 \\ 5 & 6 & 2 \\ 2 & 4 & 0 \end{vmatrix} \\
&= 3(-2)\begin{vmatrix} 1 & 2 \\ 2 & 4 \end{vmatrix} \\
&= 3(-2)(1\cdot 4 - 2\cdot 2) \\
&= 0
\end{aligned}
$$

Since the determinant of A is zero, A cannot have an inverse—by the Invertibility Criterion.

Row and Column Transformations

- The preceding example shows that, if we expand a determinant about a row or column that contains many zeros, our work is reduced considerably.
  - We don’t have to evaluate the cofactors of the elements that are zero.

Row and Column Transformations

- The following principle often simplifies the process of finding a determinant by introducing zeros into it without changing its value.

Row and Column Transformations

- If A is a square matrix, and if the matrix B is obtained from A by adding a multiple of one row to another, or a multiple of one column to another, then

$$\det(A) = \det(B)$$

Row and Column Transformations

- Find the determinant of the matrix A.

$$A = \begin{bmatrix} 8 & 2 & -1 & -4 \\ 3 & 5 & -3 & 11 \\ 24 & 6 & 1 & -12 \\ 2 & 2 & 7 & -1 \end{bmatrix}$$

  - Does it have an inverse?

Row and Column Transformations

- If we add –3 times row 1 to row 3, we change all but one element of row 3 to zeros:

$$\begin{bmatrix} 8 & 2 & -1 & -4 \\ 3 & 5 & -3 & 11 \\ 0 & 0 & 4 & 0 \\ 2 & 2 & 7 & -1 \end{bmatrix}$$

- This new matrix has the same determinant as A.

Row and Column Transformations

- If we expand its determinant by the third row, we get:

$$\det(A) = 4\begin{vmatrix} 8 & 2 & -4 \\ 3 & 5 & 11 \\ 2 & 2 & -1 \end{vmatrix}$$

  - Now, adding 2 times column 3 to column 1 in this determinant gives us the following result.

Row and Column Transformations

$$\det(A) = 4\begin{vmatrix} 0 & 2 & -4 \\ 25 & 5 & 11 \\ 0 & 2 & -1 \end{vmatrix} = 4(-25)\begin{vmatrix} 2 & -4 \\ 2 & -1 \end{vmatrix}$$

$$= 4(-25)\left[2(-1) - (-4)2\right]$$

$$= -600$$

- Since the determinant of A is not zero, A does have an inverse.

Systems of Linear Equations

A system of linear equations can be presented in different forms

$$\left.\begin{aligned} 2x_1 + 4x_2 - 3x_3 &= 3 \\ 2.5x_1 - x_2 + 3x_3 &= 5 \\ x_1 \qquad\quad - 6x_3 &= 7 \end{aligned}\right\} \Leftrightarrow \begin{bmatrix} 2 & 4 & -3 \\ 2.5 & -1 & 3 \\ 1 & 0 & -6 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 3 \\ 5 \\ 7 \end{bmatrix}$$

Standard form

Matrix form

$$|A|.|x| = |b|$$

Cramer's Rule

Linear Equations and Determinants

- The solutions of linear equations can sometimes be expressed using determinants.
  - To illustrate, let's solve the following pair of linear equations for the variable *x*.

$$\begin{cases} ax + by = r \\ cx + dy = s \end{cases}$$

Linear Equations and Determinants

- To eliminate the variable *y*, we multiply the first equation by *d* and the second by *b*, and subtract.

$$
\begin{array}{l}
adx + bdy = rd \\
bcx + bdy = bs \\
\hline
adx - bcx = rd - bs
\end{array}
$$

Linear Equations and Determinants

- Factoring the left-hand side, we get:

$$(ad - bc)x = rd - bs$$

  - Assuming that $ad - bc \neq 0$, we can now solve this equation for x:

$$x = \frac{rd - bs}{ad - bc}$$

  - Similarly, we find:

$$y = \frac{as - cr}{ad - bc}$$

Linear Equations and Determinants

- The numerator and denominator of the fractions for x and y are determinants of 2×2 matrices.
  - So, we can express the solution of the system using determinants as follows.

Cramer's Rule for Systems in Two Variables

- The linear system $\begin{cases} ax + by = r \\ cx + dy = s \end{cases}$

- has the solution

$$x = \frac{\begin{vmatrix} r & b \\ s & d \end{vmatrix}}{\begin{vmatrix} a & b \\ c & d \end{vmatrix}} \qquad y = \frac{\begin{vmatrix} a & r \\ c & s \end{vmatrix}}{\begin{vmatrix} a & b \\ c & d \end{vmatrix}}$$

- provided

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} \neq 0$$

Cramer's Rule for Systems in Two Variables

- Using the notation for determinants

$$D = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \quad D_x = \begin{bmatrix} r & b \\ s & d \end{bmatrix} \quad D_y = \begin{bmatrix} a & r \\ c & s \end{bmatrix}$$

- the solution of the system can be written as:

$$x = \frac{|D_x|}{|D|} \quad \text{and} \quad y = \frac{|D_y|}{|D|}$$

Cramer's Rule for Systems in Two Variables

- Use Cramer's Rule to solve the system.

$$\begin{cases} 2x + 6y = -1 \\ x + 8y = 2 \end{cases}$$

Cramer's Rule for Systems in Two Variables

- For this system, we have:

$$|D| = \begin{vmatrix} 2 & 6 \\ 1 & 8 \end{vmatrix} = 2 \cdot 8 - 6 \cdot 1 = 10$$

$$|D_x| = \begin{vmatrix} -1 & 6 \\ 2 & 8 \end{vmatrix} = (-1)8 - 6 \cdot 2 = -20$$

$$|D_y| = \begin{vmatrix} 2 & -1 \\ 1 & 2 \end{vmatrix} = 2 \cdot 2 - (-1)1 = 5$$

Cramer's Rule for Systems in Two Variables

- The solution is:

$$x = \frac{|D_x|}{|D|} = \frac{-20}{10} = -2$$

$$y = \frac{|D_y|}{|D|} = \frac{5}{10} = \frac{1}{2}$$

Cramer's Rule

- Cramer's Rule can be extended to apply to any system of n linear equations in n variables in which the determinant of the coefficient matrix is not zero.

Cramer's Rule

- As we saw in the preceding section, any such system can be written in matrix form as:

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix}$$

Cramer's Rule

- By analogy with our derivation of Cramer's Rule in the case of two equations in two unknowns, we let:
  - D be the coefficient matrix in this system.
  - D_{x_i} be the matrix obtained by replacing the i_{th} column of D by the numbers $b_1, b_2, \ldots, b_n$ that appear to the right of the equal sign.
- The solution of the system is then given by the following rule.

Cramer's Rule

- Suppose a system of n linear equations in the n variables $x_1, x_2, .., x_n$ is equivalent to the matrix equation $DX = B$, and $|D| \neq 0$.

  - Then, its solutions are:

$$x_1 = \frac{|D_{x_1}|}{|D|}, x_2 = \frac{|D_{x_2}|}{|D|}, \ldots, x_n = \frac{|D_{x_n}|}{|D|}$$

where D_{x_i} is the matrix obtained by replacing the i_{th} column of D by the $n \times 1$ matrix B.

Cramer's Rule for a System of Three Variables

- Use Cramer's Rule to solve the system.

$$\begin{cases} 2x - 3y + 4z = 1 \\ x \qquad\quad + 6z = 0 \\ 3x - 2y \qquad\quad = 5 \end{cases}$$

- First, we evaluate the determinants that appear in Cramer's Rule.

Cramer's Rule for a System of Three Variables

$\lvert D\rvert = \begin{vmatrix} 2 & -3 & 4 \\ 1 & 0 & 6 \\ 3 & -2 & 0 \end{vmatrix} = -38$	$\lvert D_x\rvert = \begin{vmatrix} 1 & -3 & 4 \\ 0 & 0 & 6 \\ 5 & -2 & 0 \end{vmatrix} = -78$
$\lvert D_y\rvert = \begin{vmatrix} 2 & 1 & 4 \\ 1 & 0 & 6 \\ 3 & 5 & 0 \end{vmatrix} = -22$	$\lvert D_z\rvert = \begin{vmatrix} 2 & -3 & 1 \\ 1 & 0 & 0 \\ 3 & -2 & 5 \end{vmatrix} = 13$

- Note that D is the coefficient matrix and that D_x, D_y, and D_z are obtained by replacing the first, second, and third columns of D by the constant terms.

Cramer's Rule for a System of Three Variables

- Now, we use Cramer's Rule to get the solution:

$$x = \frac{|D_x|}{|D|} = \frac{-78}{-38} = \frac{39}{19}$$

$$y = \frac{|D_y|}{|D|} = \frac{-22}{-38} = \frac{11}{19}$$

$$z = \frac{|D_z|}{|D|} = \frac{13}{-38} = -\frac{13}{38}$$

Limitations of Cramer's Rule

- In systems with more than three equations, evaluating the various determinants involved is usually a long and tedious task.
- Moreover, the rule doesn't apply if $|D| = 0$ or if D is not a square matrix.